Why Does a Detailed Rubric Hurt gpt-oss-20b?

A mechanistic interpretability investigation on BFF-Bench, following the logit-lens and causal-activation-patching method of Wang et al. 2025 (arXiv:2508.02087).

Team 10 × Kensho · AC297r · October 2026

> *A teammate noticed something strange: grading an AI judge's answers with a more detailed rubric made almost every model agree with human graders more often — except one. gpt-oss-20b got worse. This report traces that anomaly from a one-line observation to a measurable, causal mechanism inside the model — and is honest about where the trail goes cold.*

1. Confirming the anomaly is real

Before looking inside any model, the effect had to survive a basic check: is this really about the rubric, or just noise? The test used BFF-Bench's own human-annotator rubric — not something invented for this investigation — applied verbatim as grading instructions to all 8 judges in the project's model pool.

Verbatim: Appendix A.4, "Annotator Guidelines," Setlur et al., *No Free Labels* (arXiv:2503.05061)

1	If the model's final answer or recommendation disagrees with the reference, mark it "Incorrect."
2	If the model agrees with the reference but its intermediate steps disagree, also mark it "Incorrect."
3	If the model agrees with the reference, all steps agree, and it adds information that isn't obviously wrong, mark it "Correct."
4	If the model agrees with the reference, all steps agree, but the answer is slightly off from rounding, mark it "Correct."
5	If the model agrees with the reference, all steps agree, but is missing some information, judge whether that's crucial; if unsure, lean "Correct" or mark "Unsure."

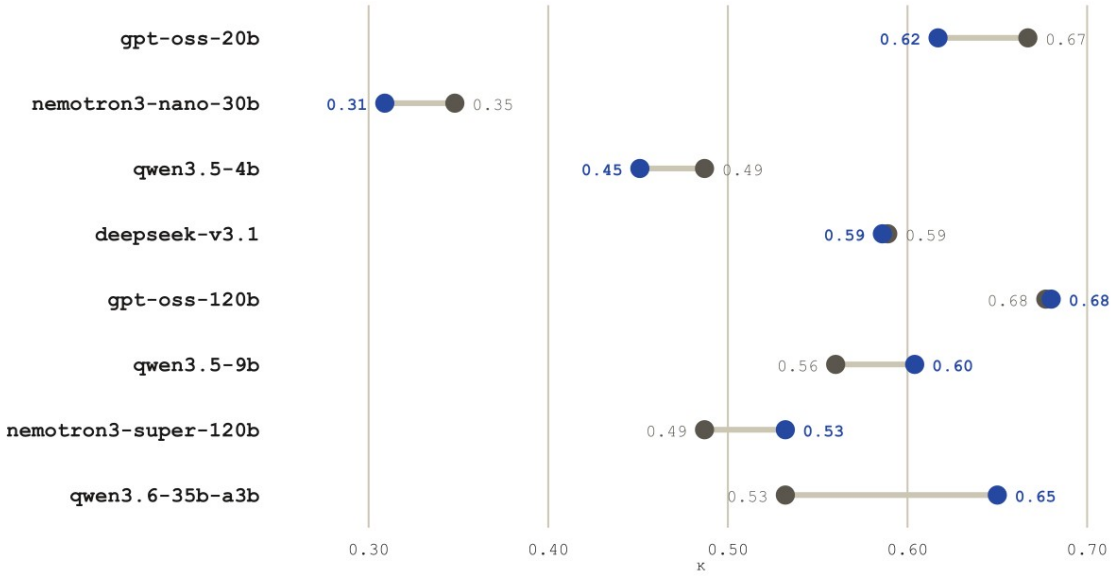


Figure 1. Agreement with human labels (Cohen's κ), no rubric (gray) vs. the real rubric (navy), all 8 judges, sorted by effect size. **gpt-oss-20b is the only judge that moves backward meaningfully**; six of eight hold steady or improve.

The anomaly held up. The next question was *why* — and the rubric's own structure gave a direct way to test it. Of its five rules, only Rule 2 asks the judge to check the solution's *steps*, not just its final answer. Removing just that one rule, and re-running the same judges:



Figure 2. Real rubric (gray) vs. the same rubric with Rule 2 removed (navy). **Removing Rule 2 alone fully restores gpt-oss-20b** (κ 0.617→0.666, matching its no-rubric baseline of 0.667); gpt-oss-120b is unchanged either way. deepseek-v3.1 moves the opposite direction — a side finding this report doesn't explain.

> **Confirmed** The effect is real, specific, and attributable: **Rule 2 — "steps must match the reference" — is what hurts gpt-oss-20b**, not the rubric in general. gpt-oss-120b, the larger sibling in the same model family, is unaffected: this is a property of the 20B checkpoint, not the gpt-oss architecture.

2. A complication: looking inside the model needed a different lens

Confirming an effect behaviorally is one thing; explaining it mechanistically is another. The

method for the second part — reading a model's internal layers via the logit lens, then testing causality with activation patching — requires a model to answer immediately, with no reasoning, so there is exactly one token position to read. That's a reasonable simplification for most purposes. For gpt-oss-20b under the real rubric, it was fatal: forced to answer instantly, **gpt-oss-20b's verdict changed on just 1 of 459 items**. The behavioral effect (κ 0.667→0.617) is real, but it evaporates the moment the model isn't allowed to think it through.

That gap turned out to be explainable, not just an inconvenience. Comparing each judge's accuracy with reasoning against its accuracy without reasoning shows how much each one actually depends on the reasoning step to get these questions right at all:

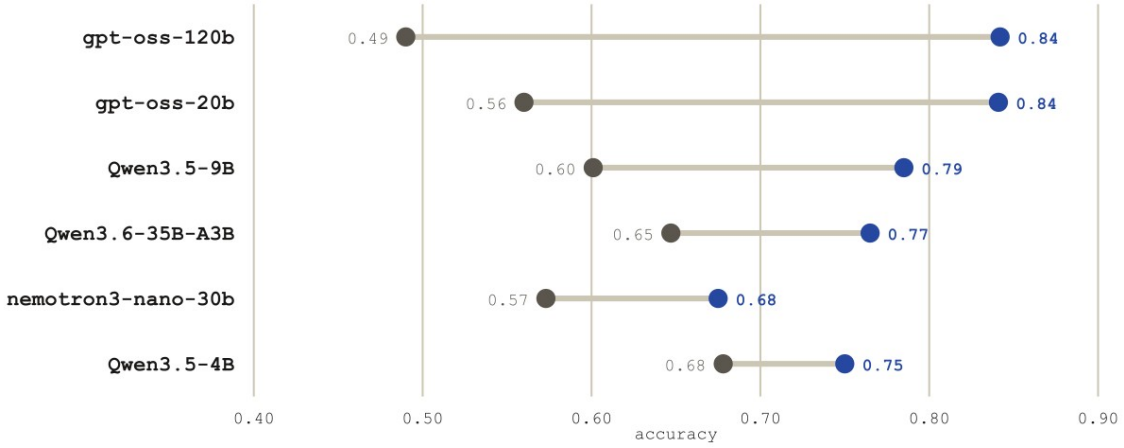


Figure 3. Accuracy with reasoning (navy) vs. without it (gray), six models tested, sorted by gap. **The gpt-oss family depends on reasoning the most** (+0.35 for the 120B model, +0.28 for the 20B model) — **exactly where the real-rubric probe found nothing to work with.** nemotron3-nano-30b and Qwen3.5-4B depend on it least (+0.10, +0.07), and gave the clearest results in either direction.

> **A real tradeoff, stated honestly** Rule 2 requires an actual comparison — do the steps match the reference? — which is a multi-step reasoning operation. A model that can't reason before answering can't perform that comparison, so the real rubric goes silent in exactly the judges that lean on reasoning the most. **The main investigation below therefore uses a broader, stricter rubric written for this purpose** instead of the real one: it stacks several bluntly-worded conditions that can bias an immediate answer directly, without reasoning, so it produces a usable signal in every model tested — including gpt-oss-20b. That makes it a more *sensitive* probe, not a more *faithful* one. Section 4 confirms the key result holds on the real rubric too.

3. What's actually happening inside the model

At every layer of the network, a technique called the logit lens reads out what the model would

answer if it had to decide right there — before the rest of the network gets a chance to process further. Tracking this layer by layer, for the items where the rubric actually changes the verdict, shows where in the network the rubric's influence takes hold.

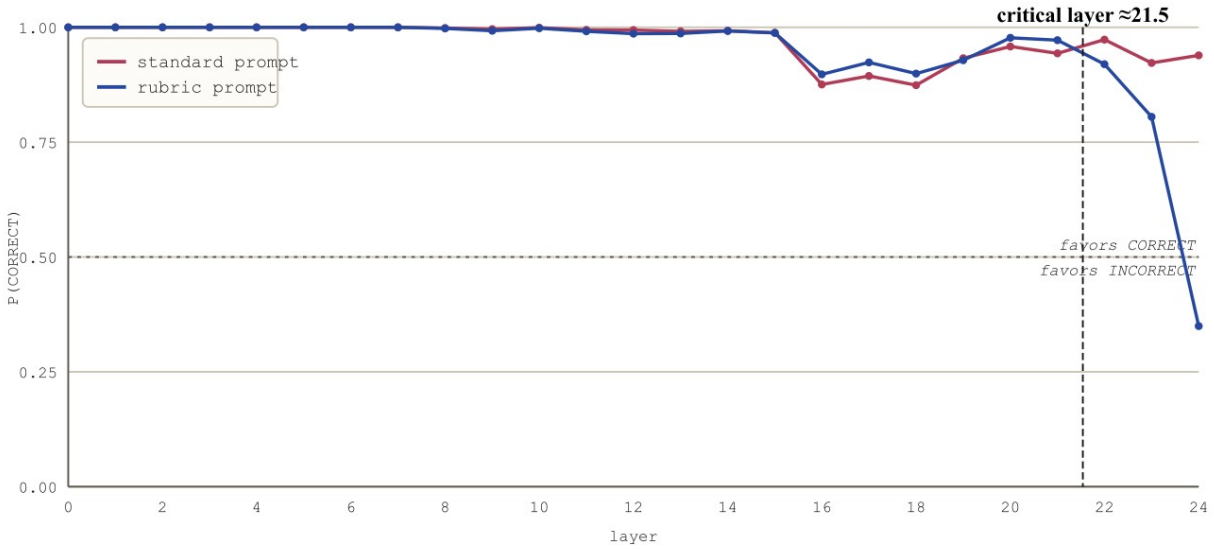


Figure 4. gpt-oss-20b, the 24 items that flip from CORRECT to INCORRECT when the rubric is added. Both curves favor CORRECT through the middle of the network; the rubric curve then drops below the decision boundary **very late** — just before the critical layer (≈22 of 24, 91% of the way through).

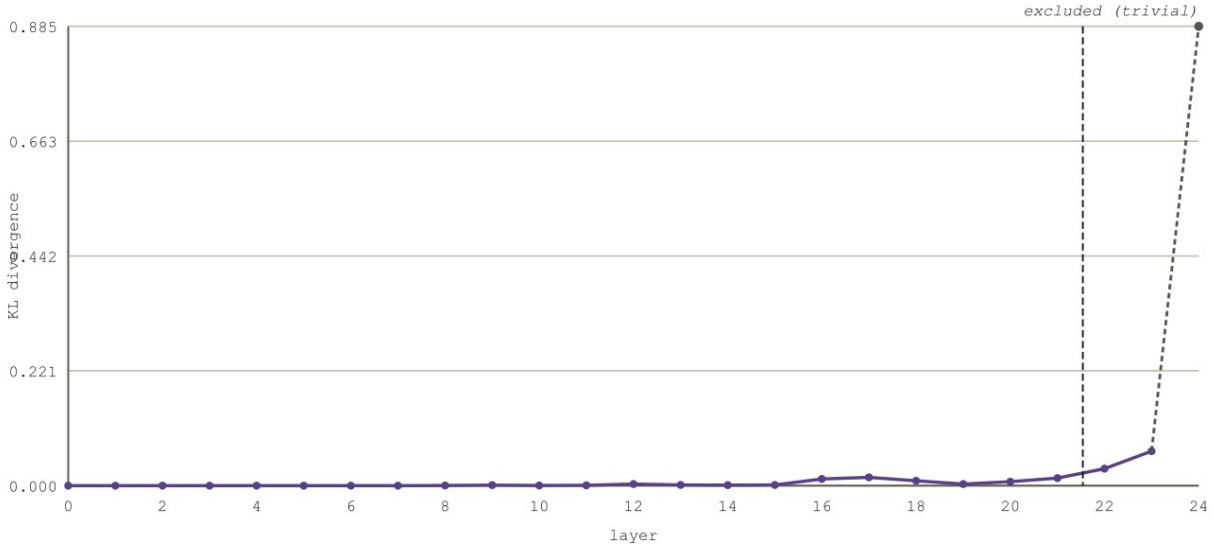


Figure 5. Layer-wise divergence between the standard and rubric-prompt output distributions, same items. The very last layer is excluded from the peak search (dashed line) because it feeds directly into the output and trivially dominates — shown here anyway, grayed out, so the exclusion is visible rather than hidden.

A late, sharp divergence like this looks like a *decision point* — a specific place where the network's internal answer flips. The only way to know if it's actually doing the work, rather than just correlating with the final answer, is to intervene: swap the hidden state at that exact layer between the two prompt conditions and see if the verdict follows.

> **Causal result** Patching gpt-oss-20b's critical layer (a 3-layer window centered on it) **recovers the correct verdict in 81% of flipped items** when the "good" activation is substituted in, but only **flips a correct run to incorrect 5% of the time** in reverse. That asymmetry — easy to break one way, hard to break the other — is the signature of a **near switch-like gate**, not a gradual shift.

Having found a clean, causal mechanism in one model, the natural next question is whether it generalizes. It doesn't — not in the way one might hope, but informatively. Five more models were tested the same way:

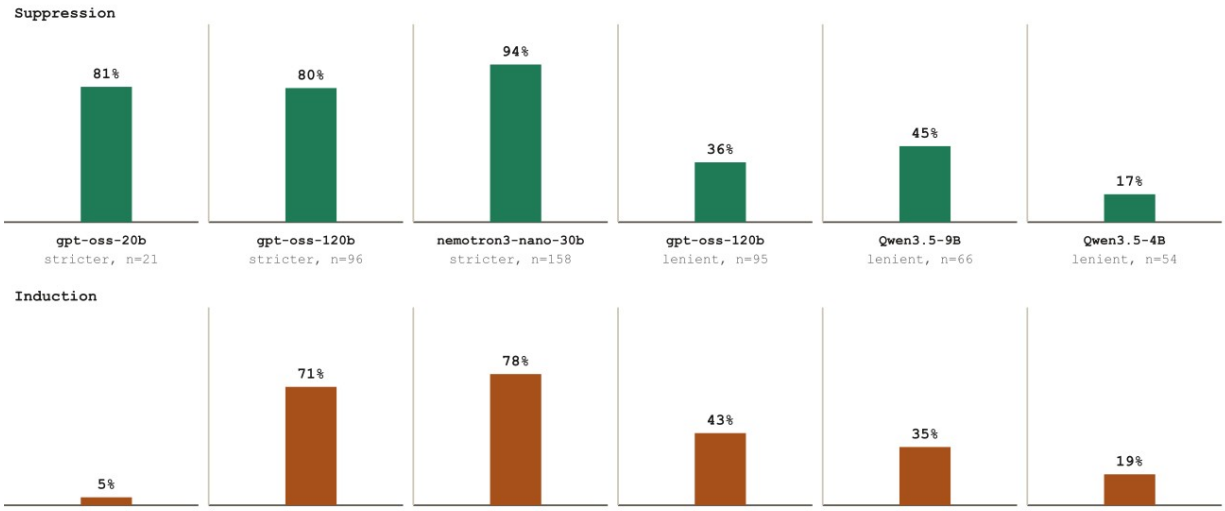


Figure 6. Causal patching results across six judges (researcher-written rubric throughout). **gpt-oss-20b**: a one-way gate (recovers almost always, rarely breaks). **gpt-oss-120b and nemotron3-nano-30b, stricter-direction flips**: strongly causal in *both* directions — a powerful, reversible pivot, not a one-way gate. The lenient-direction flips (gpt-oss-120b, both Qwen sizes) are weaker and symmetric — a softer, distributed shift.

> **Key insight** There isn't one mechanism that explains "detailed rubrics change judges' verdicts." There are at least three distinct causal signatures — a one-way gate, a reversible pivot, and a diffuse shift — occupying the same behavioral territory, in different models.

4. Checking the headline result on the real rubric

Every result so far used the broader, researcher-written rubric, adopted because the real one goes silent for reasoning-dependent judges. That leaves an open question: does any of this still hold when the real Appendix A.4 rubric is tested the same way, in the judges that *can* respond to it without reasoning? Four of the five non-gpt-oss-20b judges tested do show a large, usable flip set under the real rubric itself.

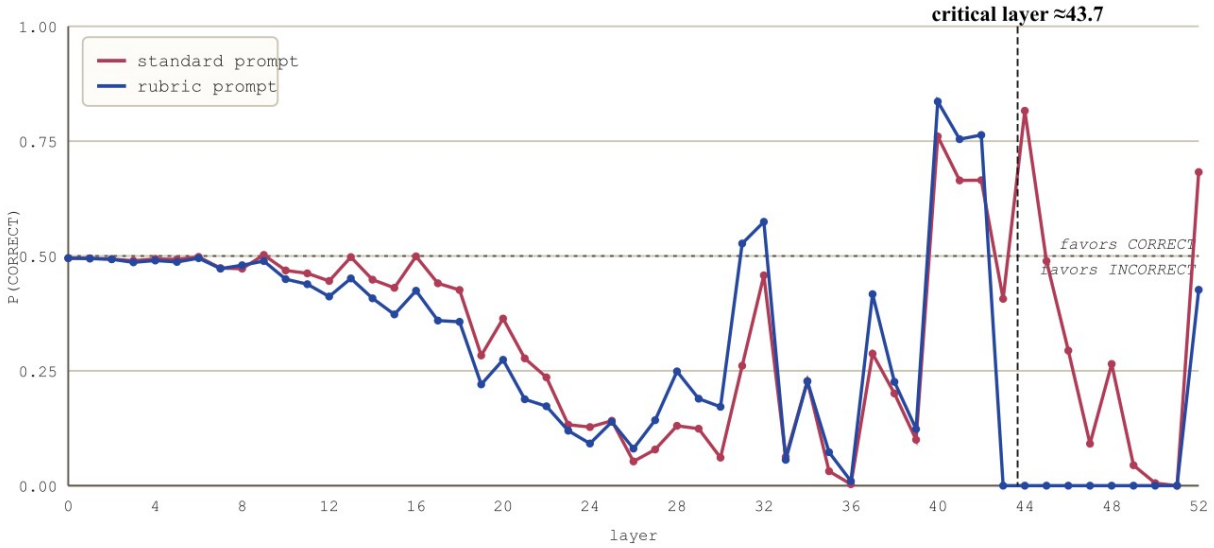


Figure 7. nemotron3-nano-30b, the 180 items that flip CORRECT→INCORRECT under the **real** Appendix A.4 rubric — nearly identical in size to its researcher-rubric flip set.

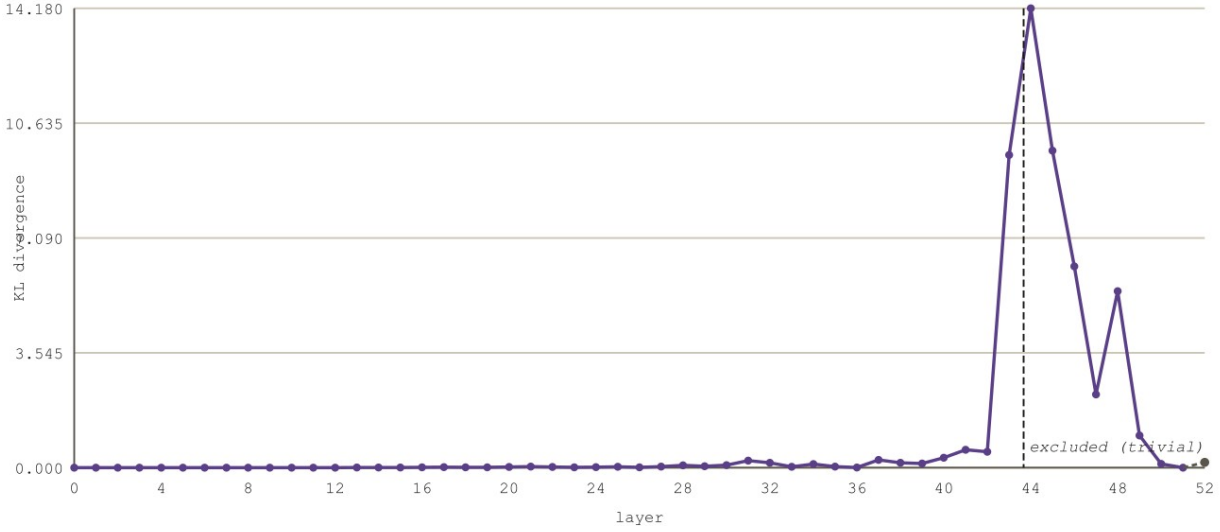


Figure 8. Layer-wise divergence, same real-rubric flip set. Critical layer ≈44 of 52 (85% depth) — also very late, like gpt-oss-20b's.

> **The strongest result in this study** Patching nemotron3-nano-30b's critical layer under the **real** rubric recovers the correct verdict **99% of the time**, and breaks a correct run **95% of the time** in reverse — a near-total, almost fully reversible pivot, on the actual rubric that produces the real-world behavioral effect. This is not a proxy result from a stand-in rubric; it's the mechanism behind the real phenomenon, found in a different judge than the one that motivated the search.

Not every judge's story is this clean. Qwen3.5-4B shows an equally large, completely one-sided behavioral shift under the real rubric (76 items, zero reversed) — but patching its own critical layer changes nothing at all:

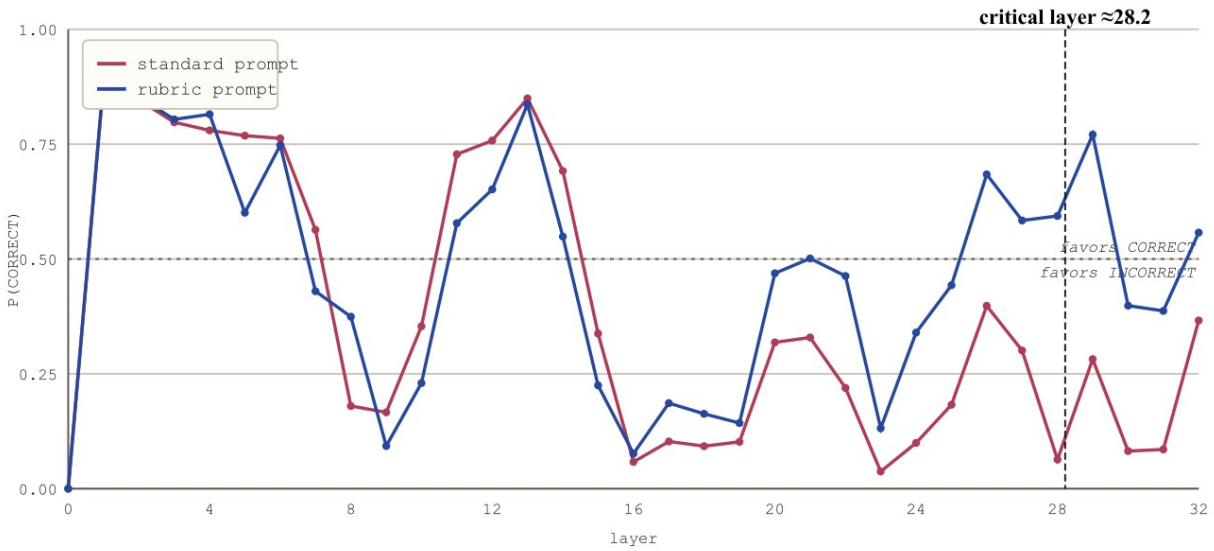


Figure 9. Qwen3.5-4B, 76 items flipping INCORRECT→CORRECT under the real rubric, zero the other way — a textbook-clean behavioral signal.

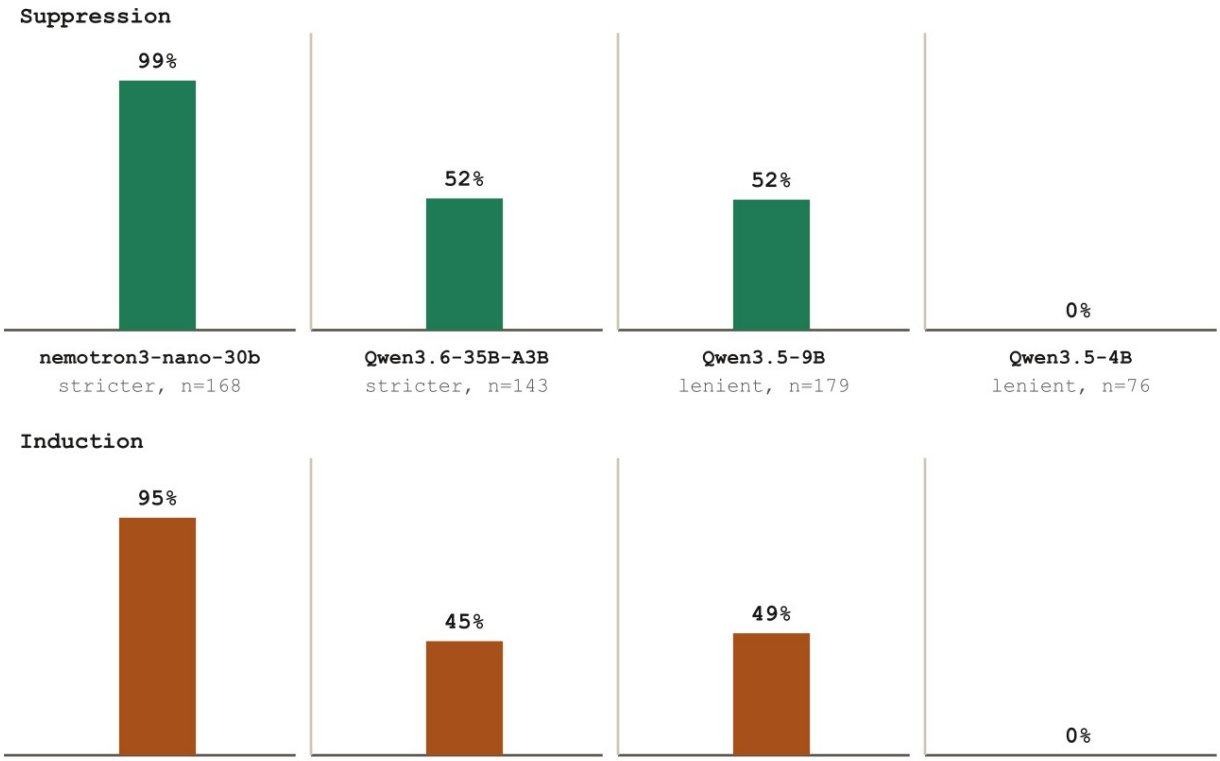


Figure 10. Causal patching, real rubric. nemotron3-nano-30b's near-total effect (99%/95%) contrasts sharply with **Qwen3.5-4B's zero effect (0%/0%)** at its own critical layer, despite an equally clean behavioral flip. Qwen3.6-35B-A3B and Qwen3.5-9B land in between (≈50%, symmetric) — two different model families converging on the same partial, distributed pattern.

> **Why this matters** A clean, one-sided behavioral flip does **not** guarantee a localized mechanism. Qwen3.5-4B's rubric sensitivity is real and measurable in its behavior, but it is not concentrated at any single layer — it's spread across the network in a way this method cannot pin down.

5. The gap that remains

The model that started this entire investigation is also the one this method explains least. Under the real rubric, forced into an immediate answer, gpt-oss-20b's verdict barely moves — 1 flip out of 459 items, far too few to build a curve or run a causal test on. Figure 3 offers a plausible reason why: gpt-oss-20b depends on reasoning more than almost any other judge tested, so a method that strips reasoning away is close to the worst possible lens for this specific model.

> **Not yet shown** A mechanism for gpt-oss-20b's own real-rubric effect specifically. Its κ drop (0.667→0.617, Figure 1) is real and reproducible, but explaining it mechanistically likely requires letting the model reason first, then reading its internal state at the final answer token — a meaningfully different method from the one used here, and not yet built.

Summary

MODEL	REAL-RUBRIC FLIPS	SUPPRESSION	INDUCTION	READING
gpt-oss-20b	1 / 459	—	—	Unexplained — too few flips to test
nemotron3-nano-30b	180	99%	95%	Near-total, reversible pivot
Qwen3.6-35B-A3B	143	52%	45%	Moderate, distributed
Qwen3.5-9B	184	52%	49%	Moderate, distributed
Qwen3.5-4B	76	0%	0%	Diffuse — no localized mechanism

Flip counts and causal effects for Qwen3.6-35B-A3B, Qwen3.5-9B, and Qwen3.5-4B are under the real Appendix A.4 rubric (§4); the researcher-written rubric (§3) was used only where the real rubric's signal was too thin, i.e. for gpt-oss-20b and as a first pass before §4's confirmation.

The story in one sentence: a single teammate's observation about one model turned out to be the visible edge of several distinct internal mechanisms, ranging from a near-binary switch to a fully diffuse shift, and the model that started the search is, so far, the one whose mechanism still can't be seen.

Team 10 × Kensho, AC297r. Data: BFF-Bench / kensho/VERDICTS (No Free Labels, arXiv:2503.05061). Method: Wang et al. 2025, arXiv:2508.02087. Interactive version with all models and full-detail figures available as a companion page.